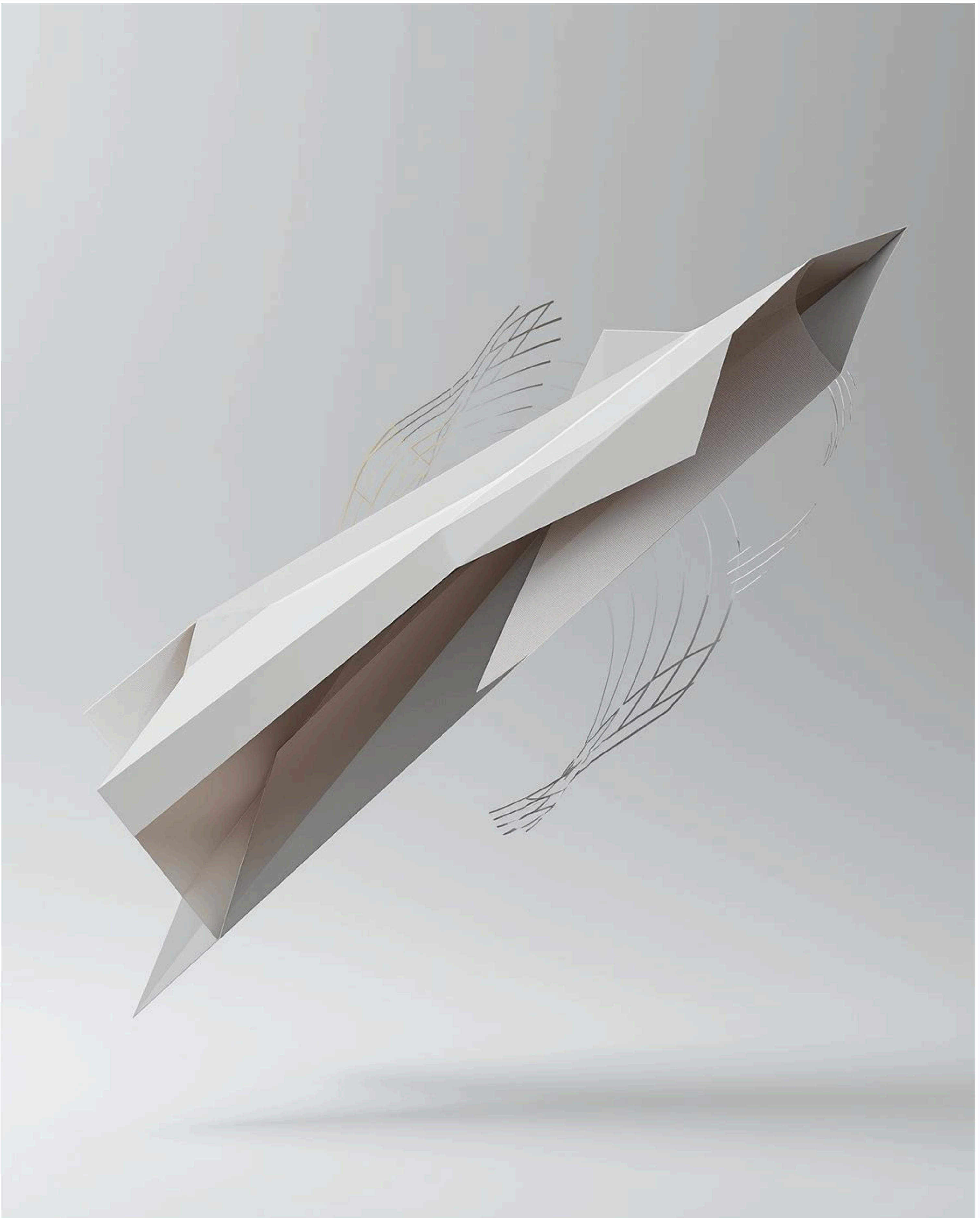

Credits: Canva

Optimal Foil Geometry for Flow Energy Harvesting

Presented by:
Chua Jun Hao Thaddeus [U2223891H]

December 8, 2025, 10:30 AM, LHN-TR+06

Background

Background

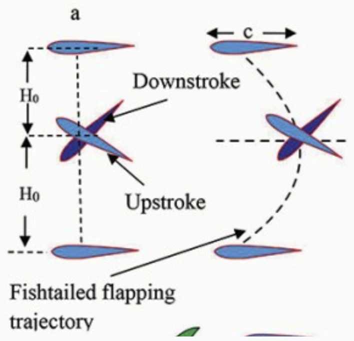
Purpose: Why this study?

(Johnston)

Global demand for **renewable / sustainable energy** sources

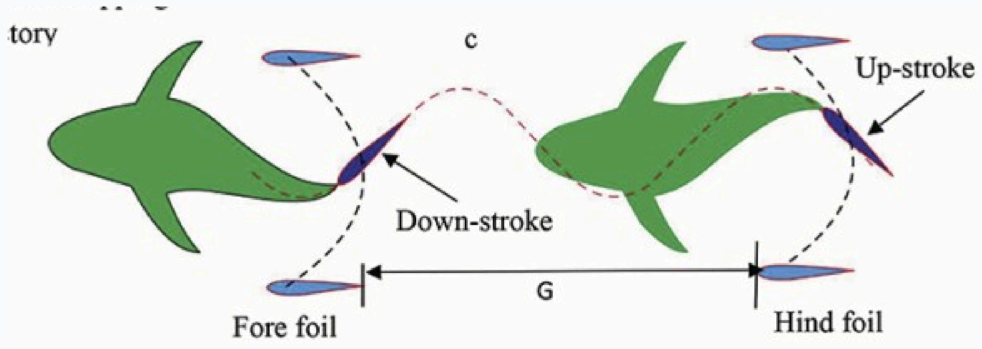
(Canva)

Underutilisation of **low-speed flow** potential



(Liebendorfer)

Importance of **flapping-foil** technology – complements turbine weaknesses

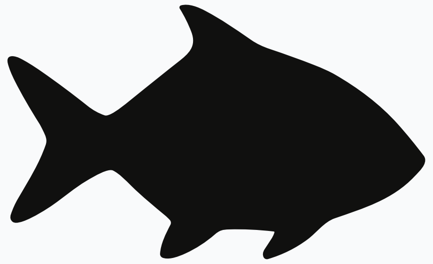


(Liebendorfer)

Increased interest in **bio-inspired** energy systems

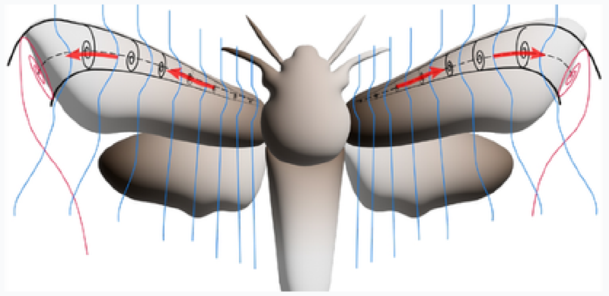
Background

Purpose: Why Flapping Foils?



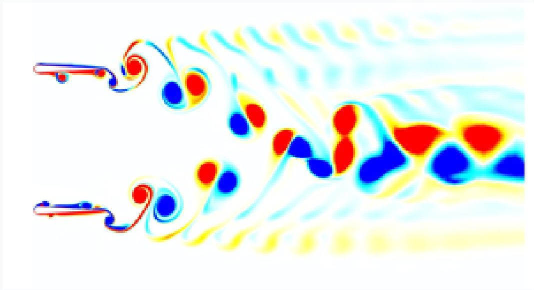
(Canva)

Ability to **mimic locomotion** observed in nature



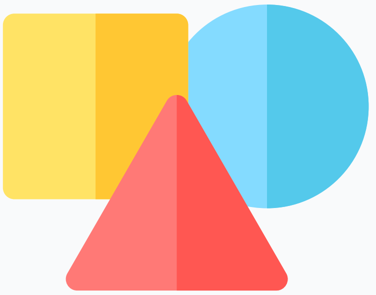
(M. De Manabendra et al.)

Advantages of bio-inspired energy systems



(Gungor, A., Khalid et al.)

Ability to **exploit unsteady vortex dynamics** to generate thrust



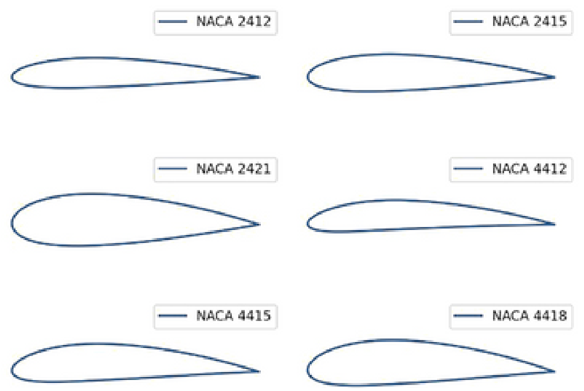
(Canva)

Strong geometric influences

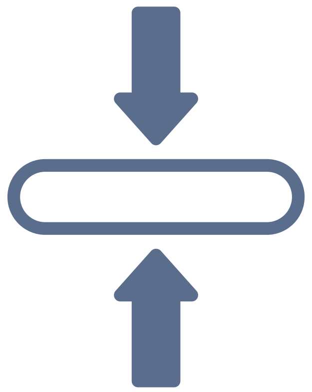
Literature Review

Literature Review

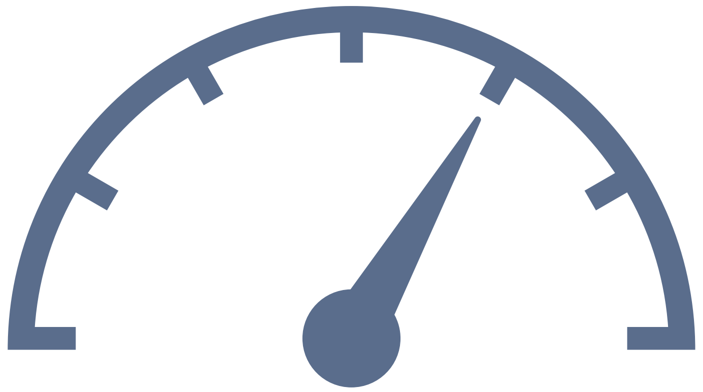
What is missing in present day investigations



(Karthikeyan and Hussain)



(Canva)



(Canva)

Foil Profile

- **Non-conventional** under-represented
- **Limited variations** in planform
- Under-studied **bio-inspired** shapes

Thickness Factor

- Not well-varied
- Often kept **fixed** or very **slightly modified**
- Limited investigations on thickness

Flow Speed

- Often tested in **higher** flow speed
- Neglects low Reynolds number
- **Limited** conditions tested

Objective

*"To investigate the combined effects of **geometry** and **thickness** on foil energy harvesting performance in low flow speed, recirculating water tunnel"*

Scope

(Canva)

Design & Preparation

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Testing & Experiment

(Canva)

Data Collection & Processing

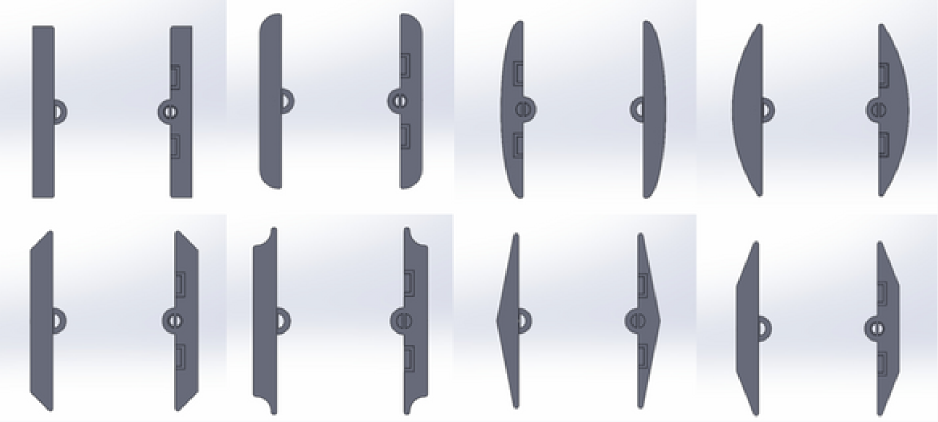
(Canva)

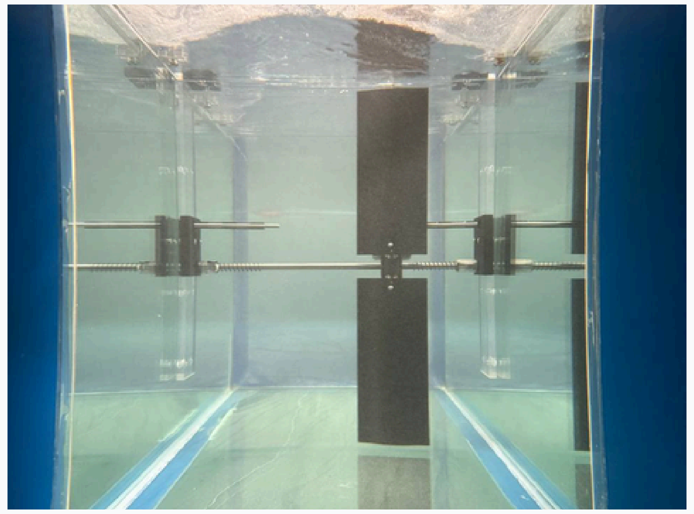
Compare Results

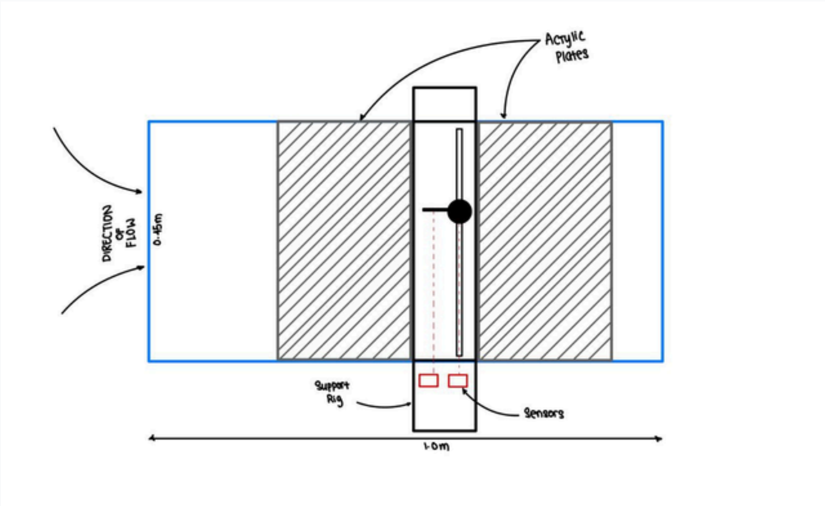
Methodology

Methodology

Quick Overview







Design Phase

Designed with CAD software, Solidworks. They were designed with **2 types** in mind: (1) only LE/TE modified, (2) full shaped planform

Testing Phase

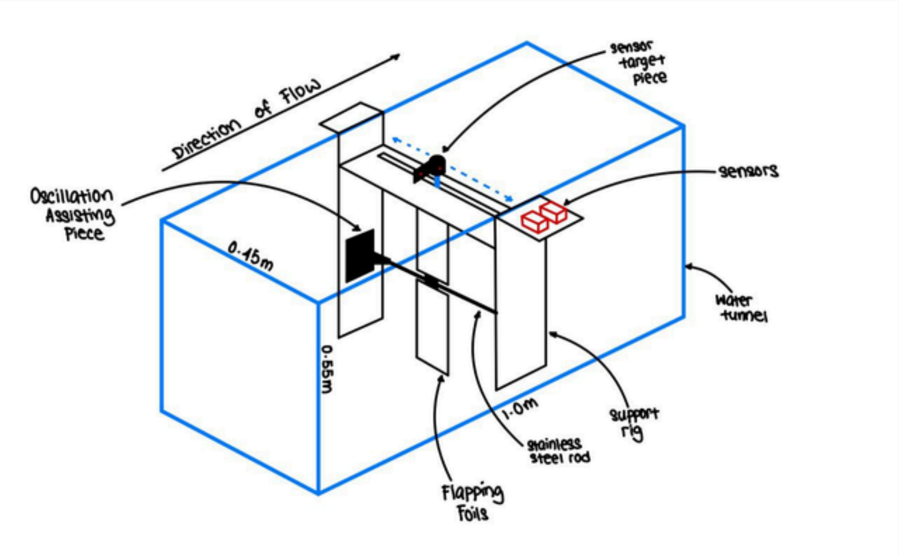
8 shapes and 1 flat, reference foils were designed & 3D printed with **PLA**. These foils are tested in a recirculating, low flow speed **water tunnel** facility

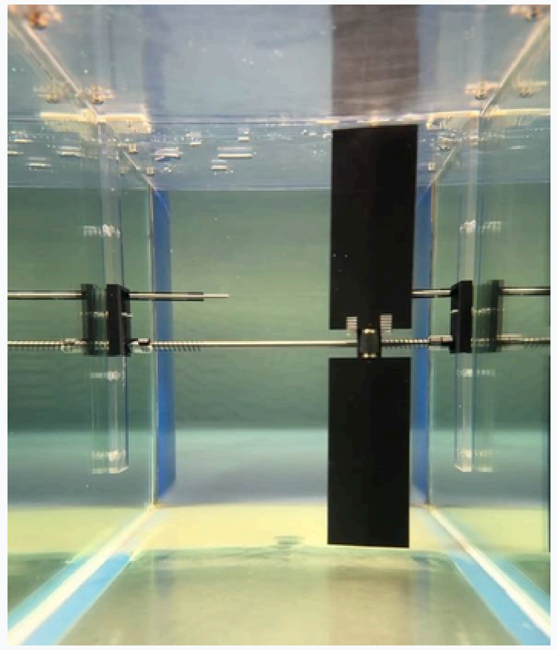
Processing Phase

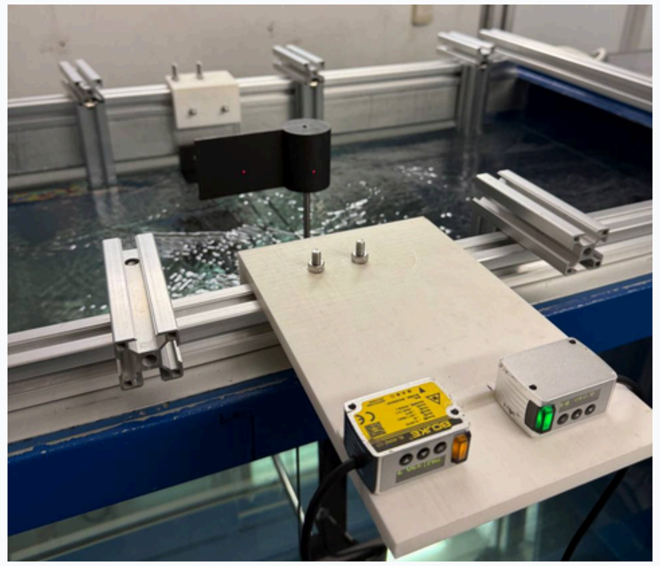
Displacement-time signal collected via laser sensors are converted using FFT into **frequency domain** and then used to obtain performance metrics

Methodology

Testing Environment and Experimental Set Up







Water Tunnel

Serves as a **controlled environment** for testing

Test section:
1.0m (L) x 0.45m (B) x 0.55m (H)

Support Rig

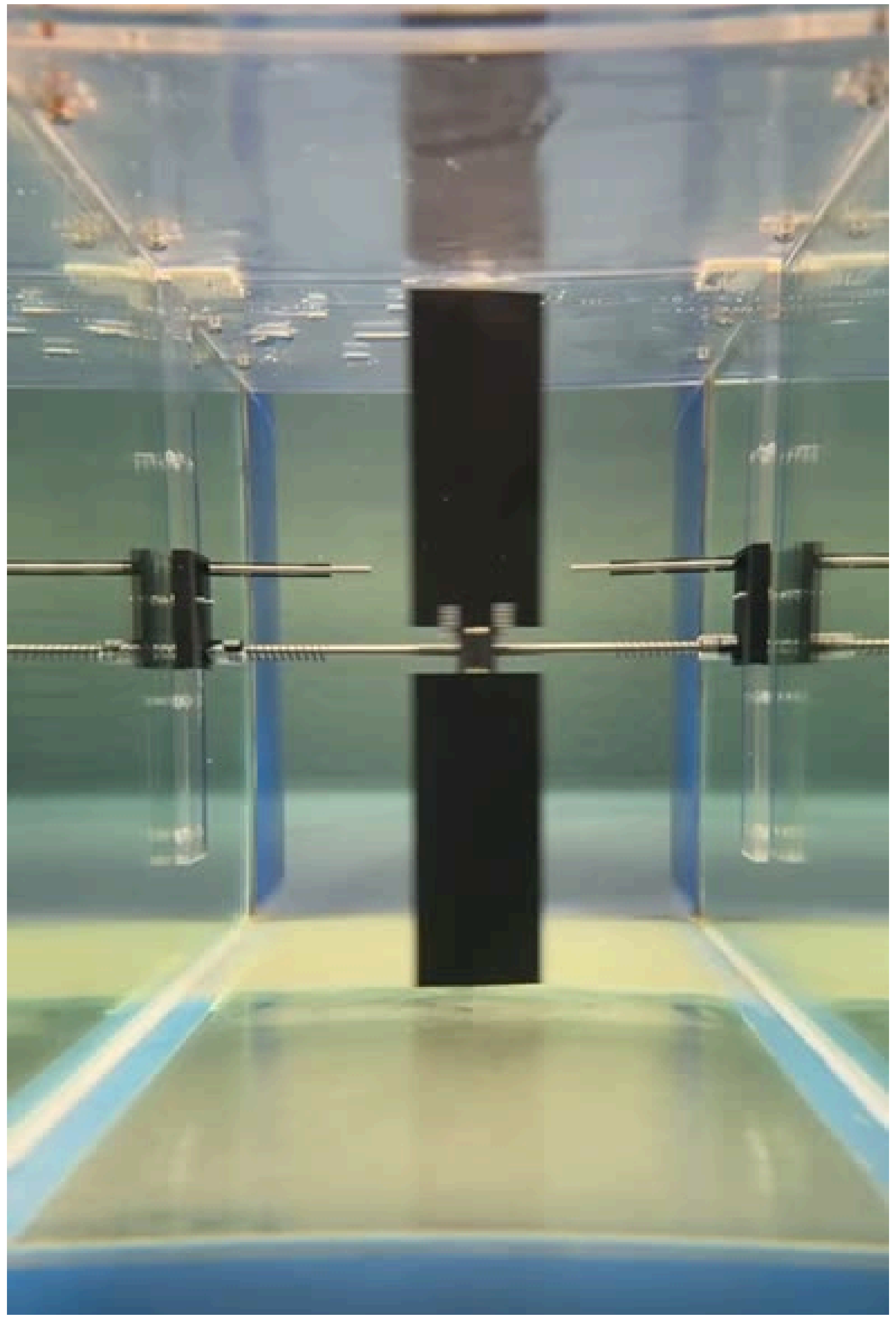
Custom **support rig** to mount the foils and sensors
Minimises movement along other axes

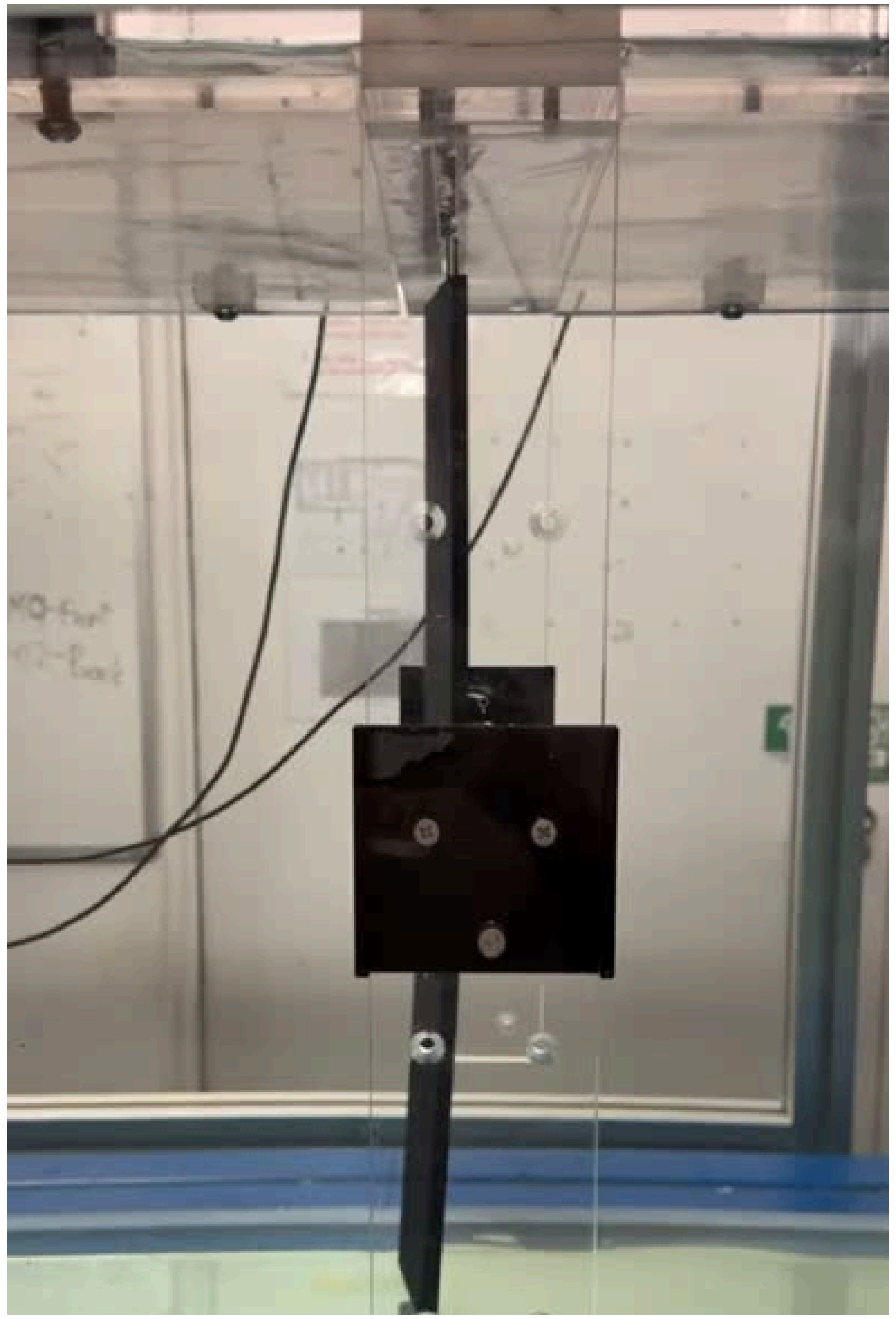
Sensors

Laser sensors (x2) are set ~50mm apart to measure the **displacement** of the foil during oscillation over time

Methodology

Testing Environment and Experimental Set Up

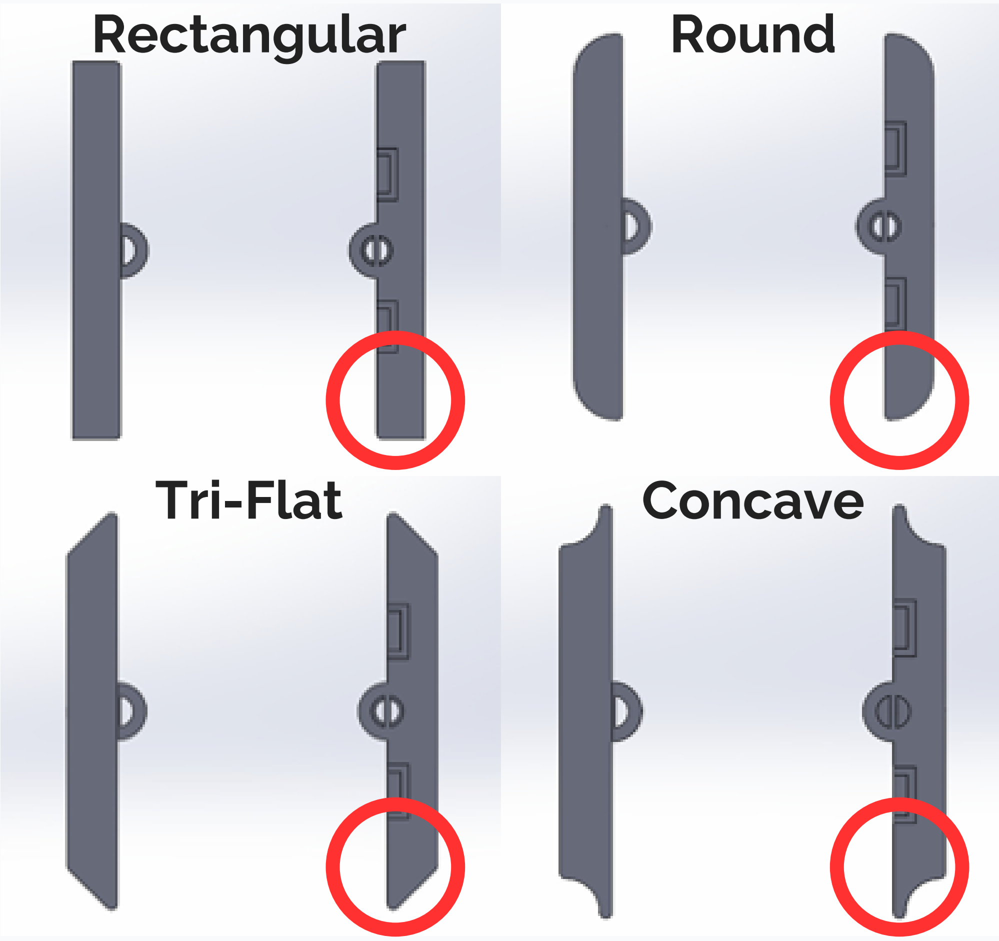




Foil Categories & Test Matrix

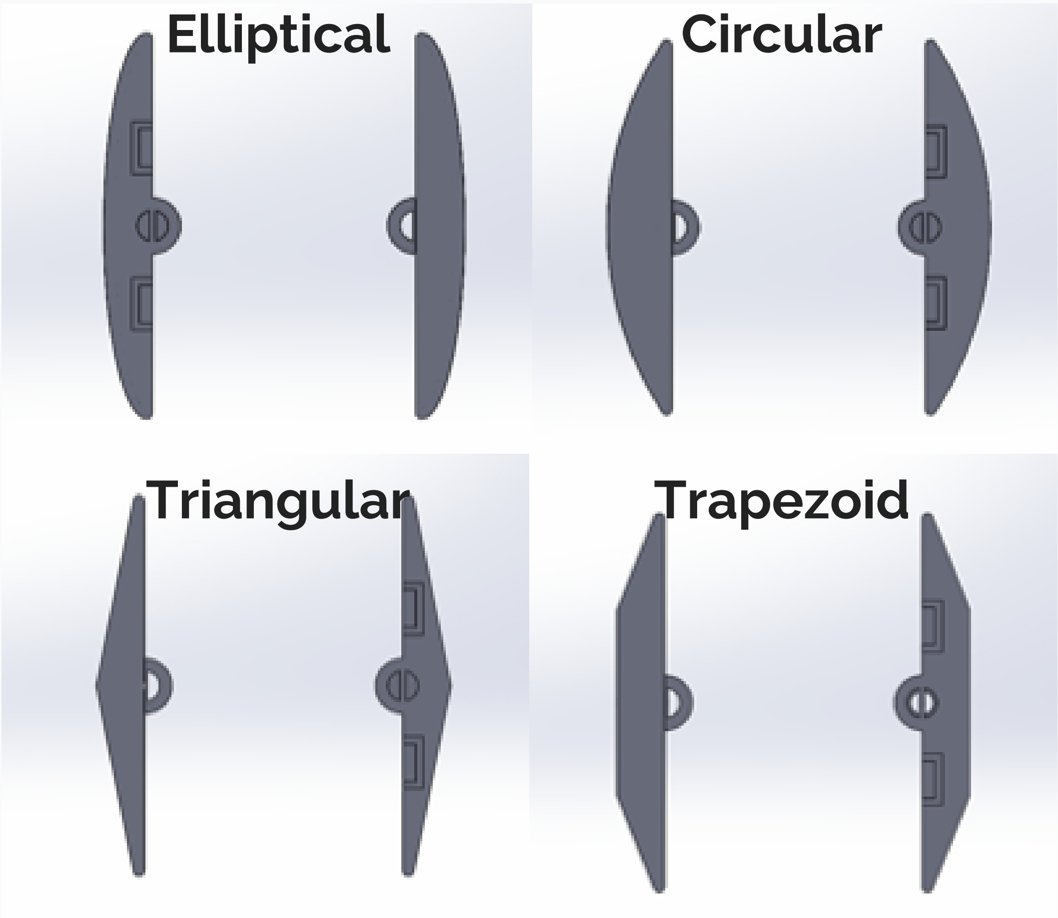
LE/TE Foil Characteristics

LE/TE foils are designed with only the **edges modified**, keeping a flat mid-section



Geometric Foil Characteristics

Geometric foils are **fully shaped** across the planform with no flat sections unless it is part of the shape (e.g. Trapezoid)

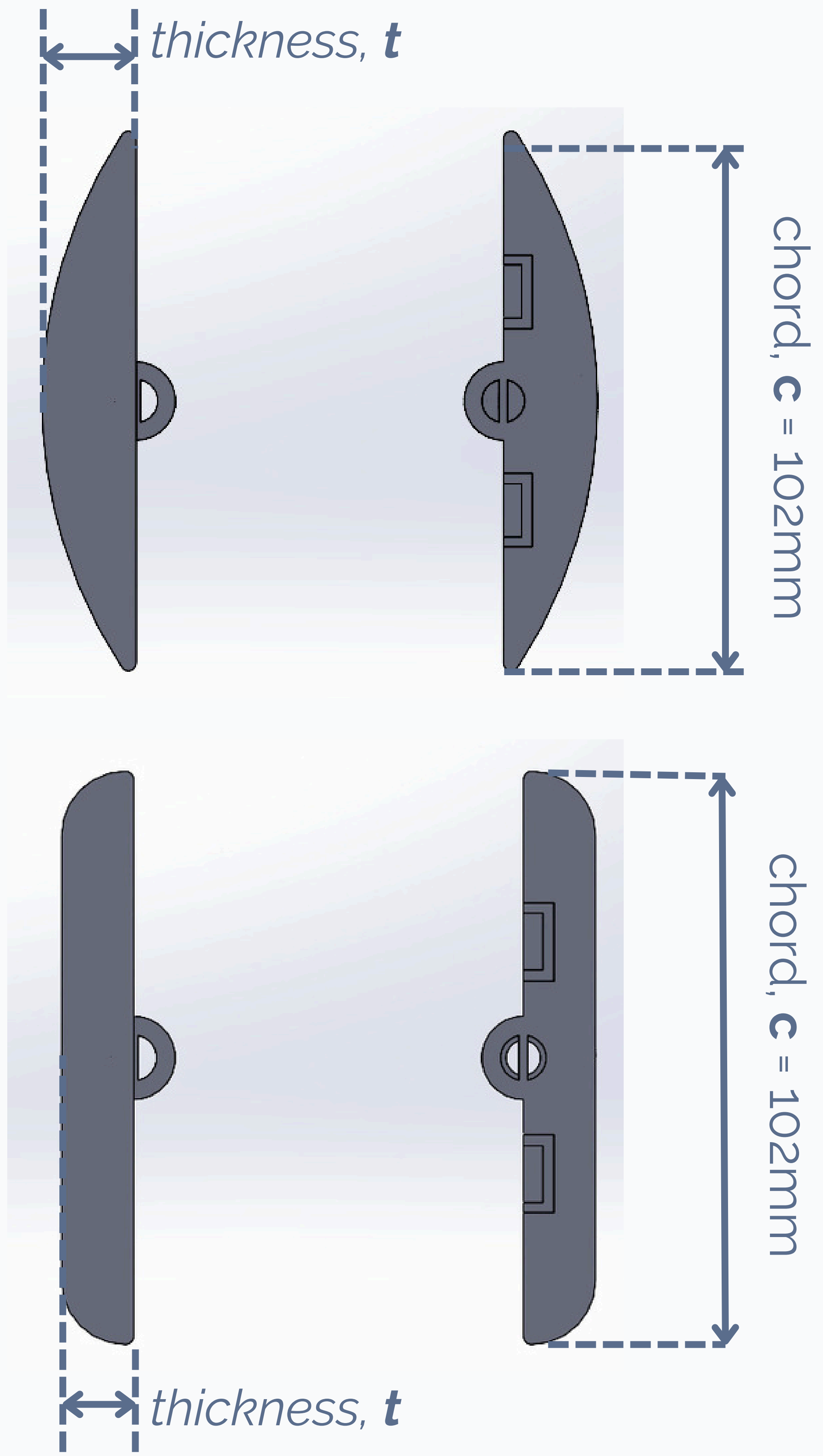


Foil Categories & Test Matrix

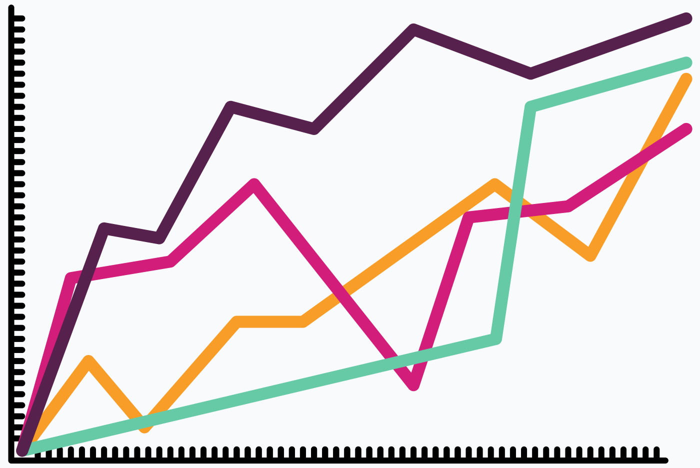
Thickness Variation Impact

Thickness plays an important role in affecting a foil's performance in testing.
4 different thicknesses are tested for each shape

5mm, 10mm, 15mm, 20mm
chord = 10.2cm = 102mm
thickness to chord ratio = **t/c**



Data Processing



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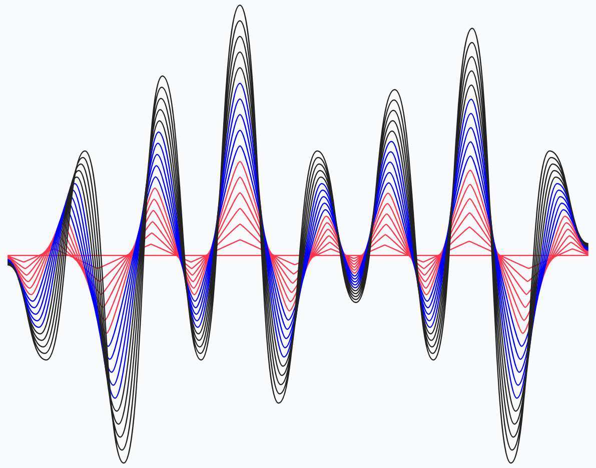

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Displacement-Time Data

Data was captured in **displacement-time signals**, but not as ideal for comparing results

FFT

Fast Fourier Transform (FFT) converts the time-based data into the **frequency domain** allowing further computation of performance metrics

Performance Metrics

Strouhal Number, Power, Efficiency

Force Calculation

$$F_L = m\ddot{h} + c_h\dot{h} \xrightarrow[\frac{1}{2}\rho U_\infty^2 bc]{\div} C_L$$

Efficiency Calculation

$$\eta = \frac{P}{\frac{1}{2}\rho U_\infty^3 bA}$$

Power Calculation

$$P = F_L\dot{h} = FU_\infty \xrightarrow[\frac{1}{2}\rho U_\infty^3 bc]{\div} C_P$$

Strouhal Number Calculation

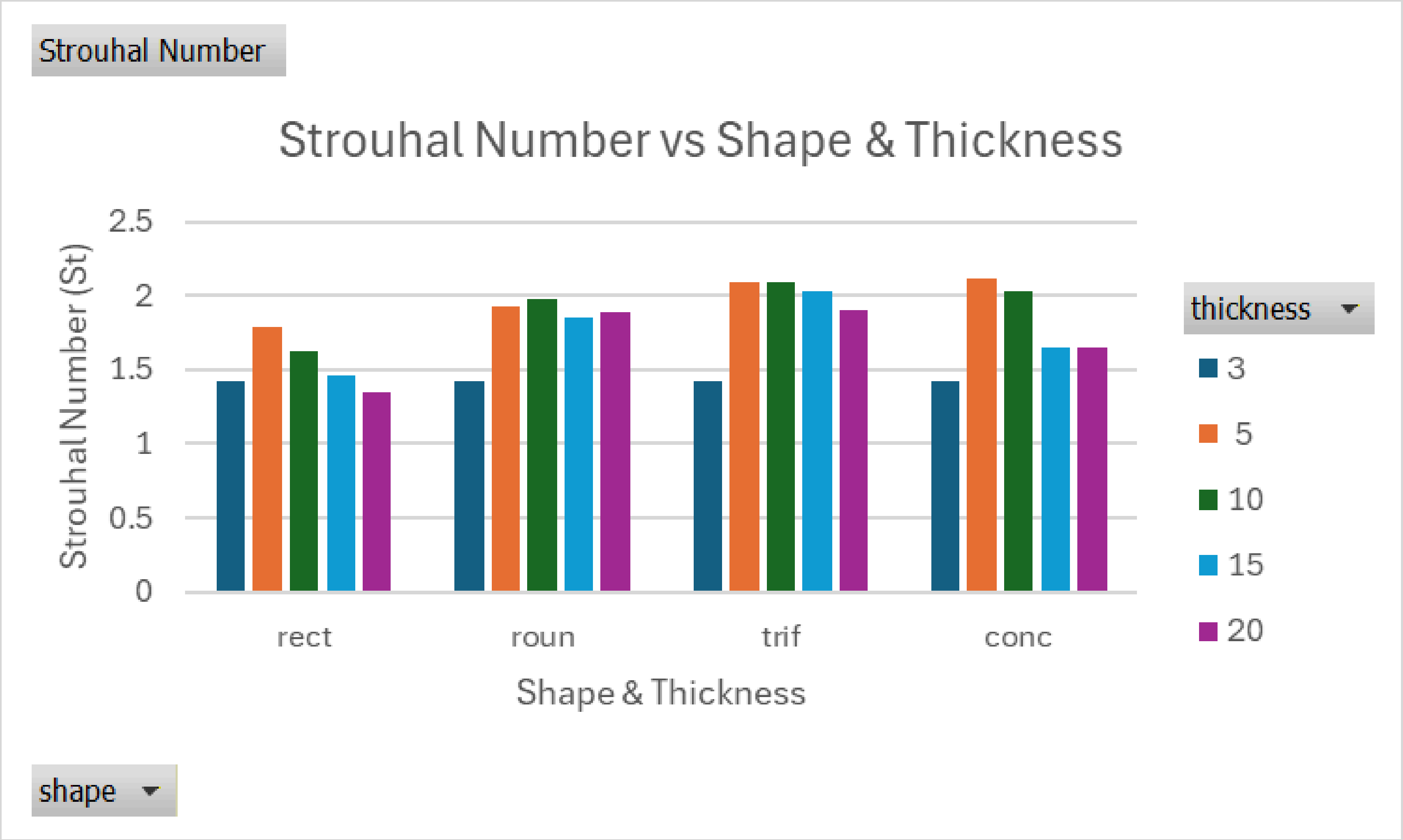
$$St = \frac{fA}{U_\infty}$$

Results

LE/TE Category

LE/TE Results

Strouhal Number



Rectangular

Lowest Strouhal numbers across all thickness

Steady decreasing trend with thickness

Round

Most consistent Strouhal numbers

Ideal at **moderate thickness**

Seems to have a **single ideal thickness**

Tri-Flat

2nd highest peak

Remains **largely consistent** across all thickness

Drops slightly at higher thickness

Concave

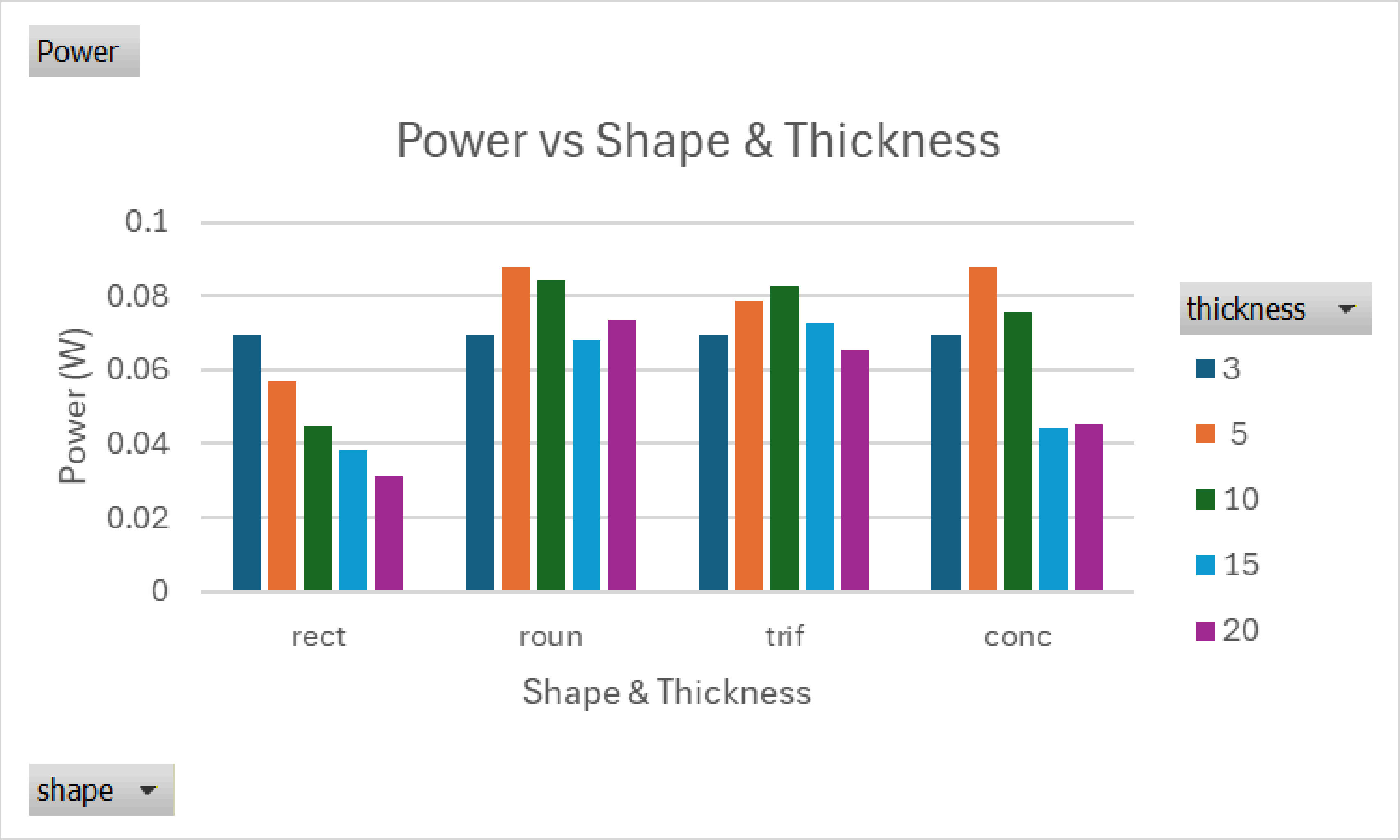
Found to have the **highest Strouhal Number** Peak

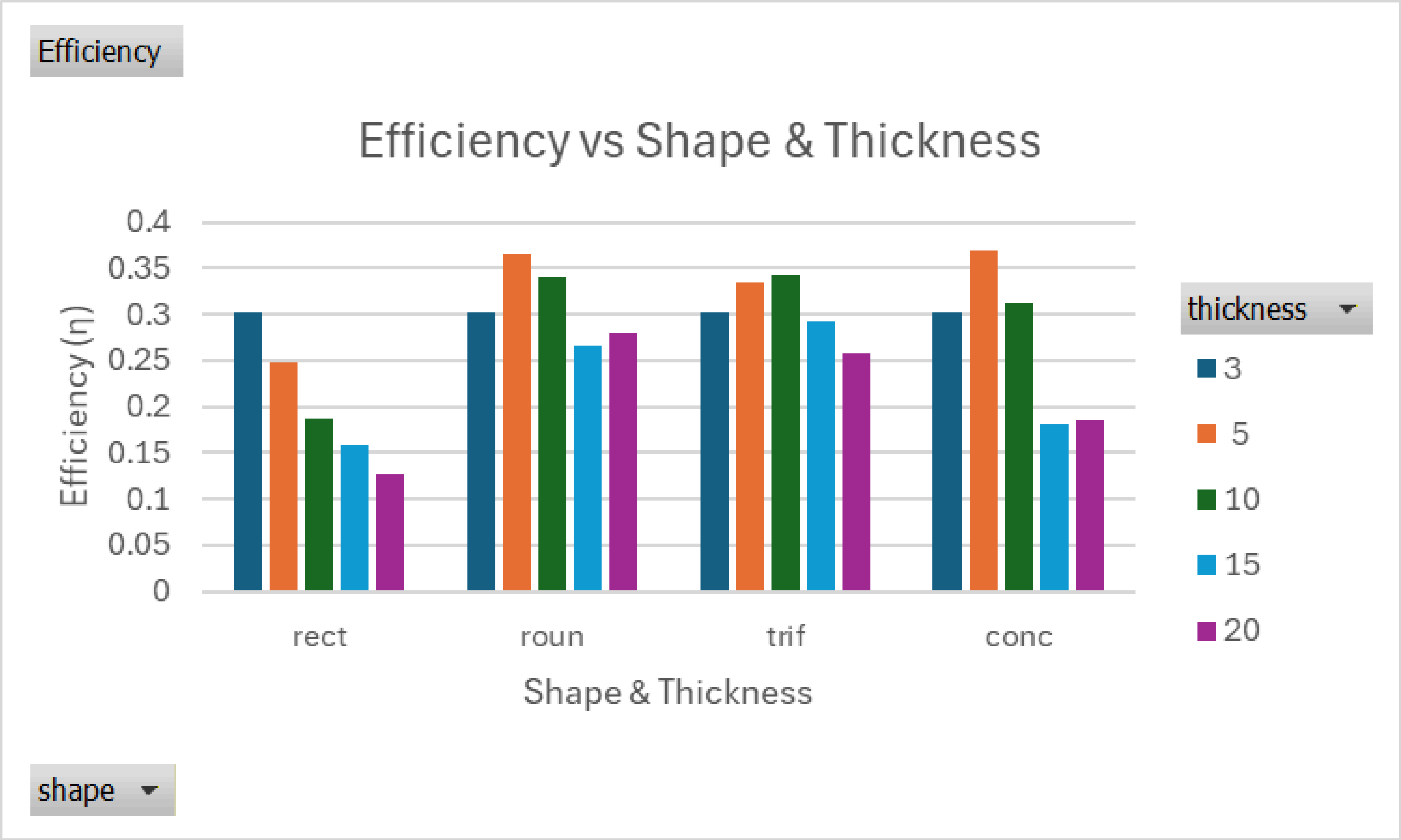
Drops significantly at higher thickness

Remains consistent at high thickness

LE/TE Results

Power & Efficiency





Rectangular

Starts off **below baseline**
Continually decreases
Poor power output observed

Round

Peaks @5mm
Steady decline over thicknesses
Rises slightly @20mm

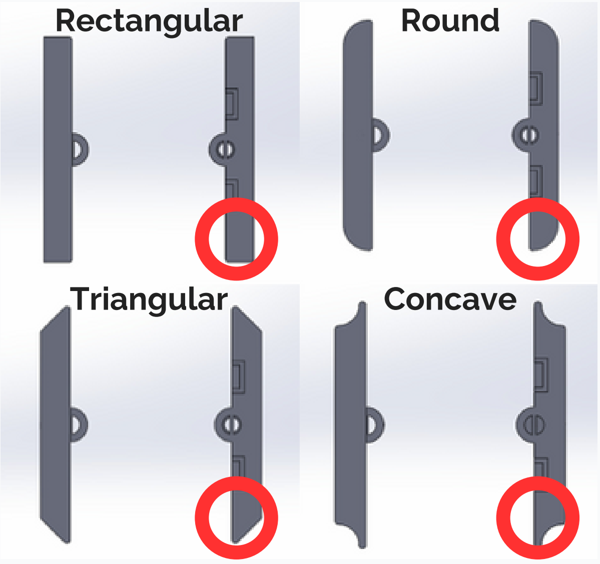
Tri-Flat

Peaks at moderate thickness
Ideal thickness **@10mm**
Power falls off at other thicknesses

Concave

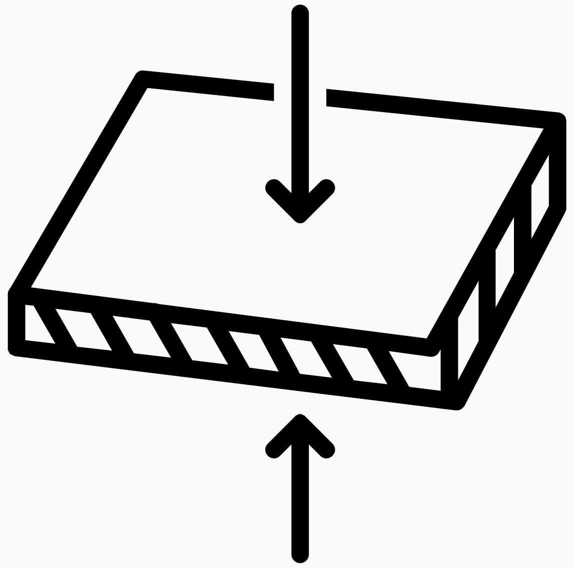
Highest power output @5mm
Highest Efficiency (~37%)
Drastic drop at higher thicknesses

Summary of Findings



Shapes

Best performing shapes were **concave and round** foils.
Blocky, highly angular shapes (Rect) are **highly ineffective**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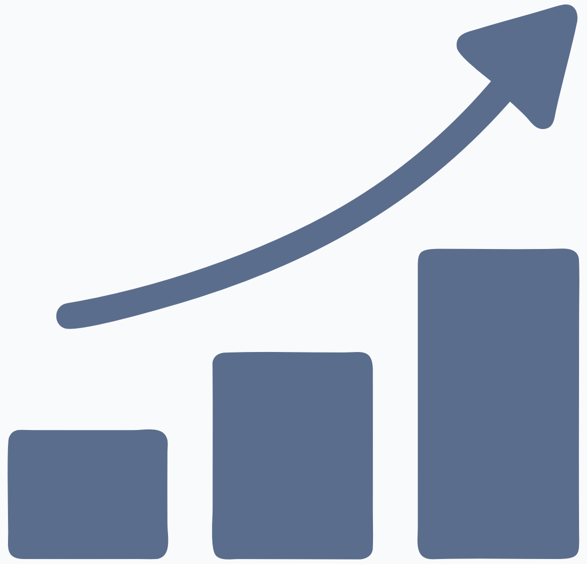
Thickness

Optimal thickness range **5-10mm**
Higher thicknesses generally negatively impacts performance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Trends

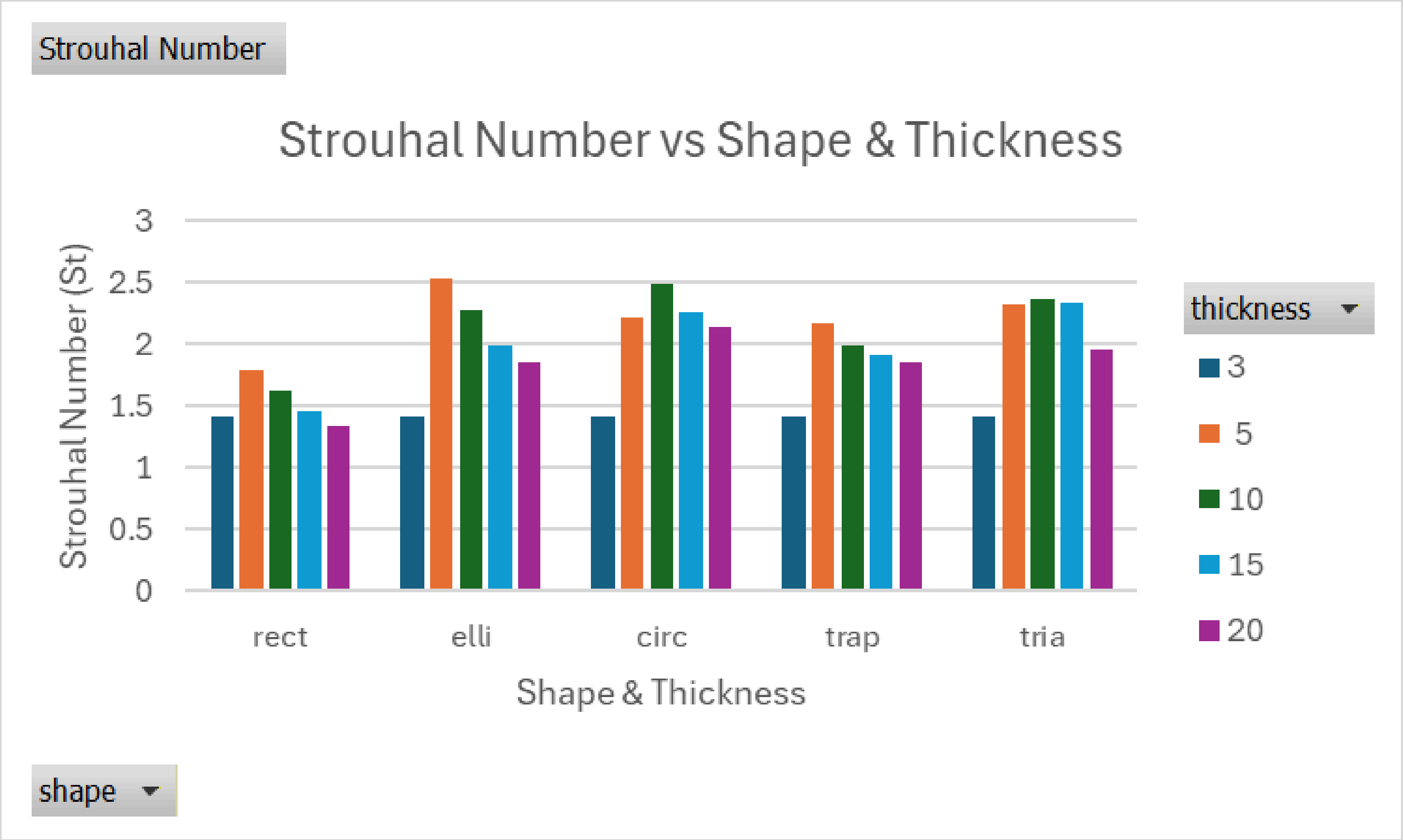
Similar trends observed across all performance metrics
Shows **strong correlation** between Strouhal number, Power and Efficiency

Results

Geometric Category

Geometric Results

Strouhal Number



Elliptical

High Peak, St = 2.53
Gradual decrease with thickness
Remains above baseline

Circular

Ideal thickness @ 10mm
Well above baseline for all thicknesses
Relatively consistent

Trapezoidal

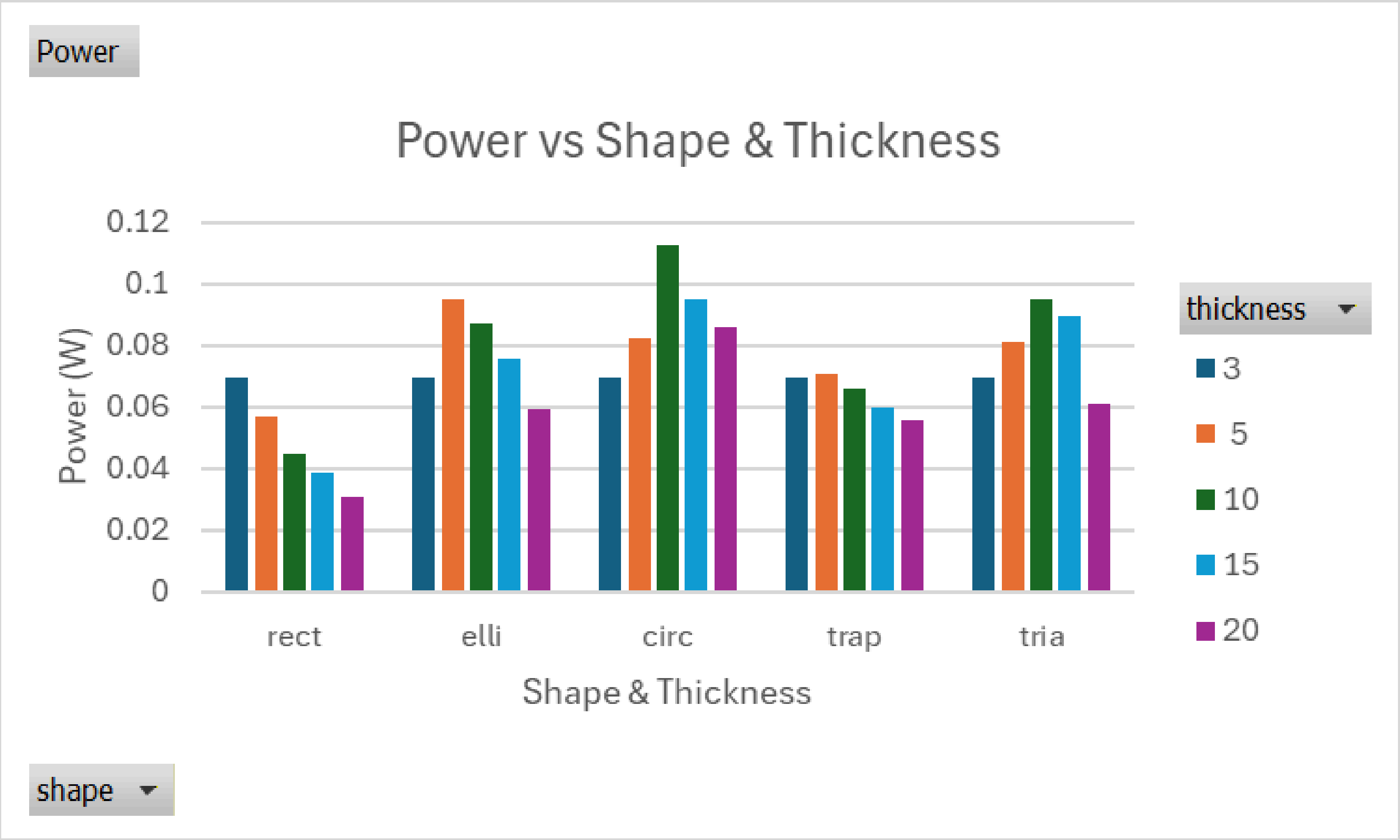
Lowest values in category
Highest at lower thickness
Slight decrease with increasing thickness

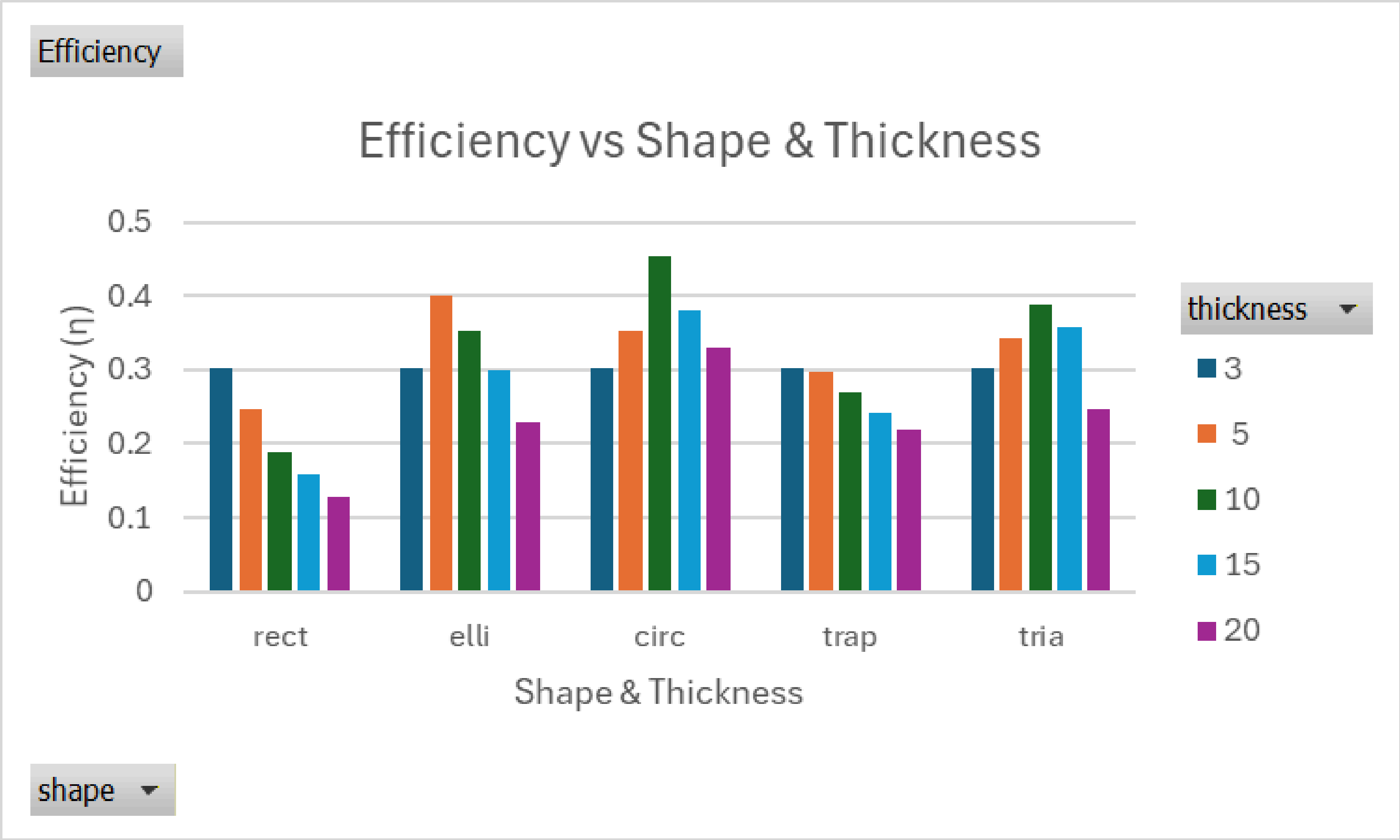
Triangular

Very consistent at moderate thicknesses
Drops significantly @20mm
Perform well over a wider range of thicknesses

Geometric Results

Power & Efficiency





Elliptical

Relatively high power output
Displays good performance
Drops @ 20mm

Circular

Highest power output (0.11W)
Significantly higher @ 10mm
Highest efficiency (~45%)

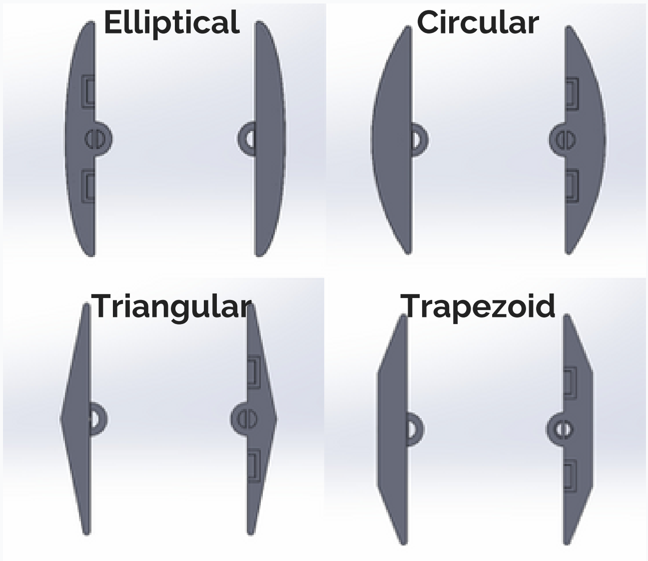
Trapezoidal

Lowest power output
Slightly above baseline @5mm
Steadily drops above this thickness

Triangular

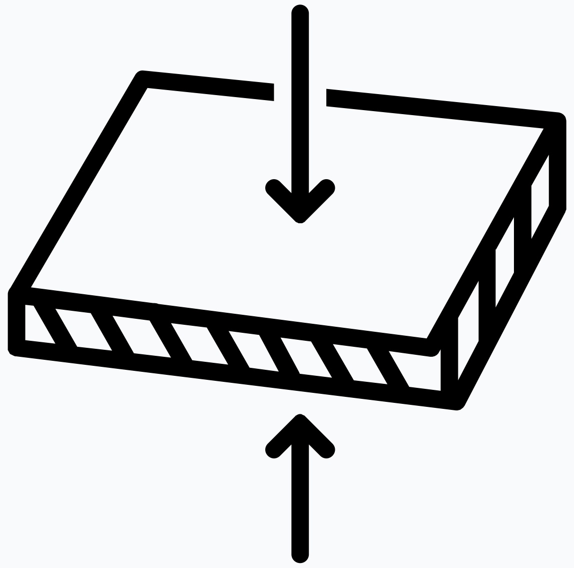
Consistent between 10-15mm
Drops drastically @ 20mm
Generally performs above average

Summary of Findings



Shapes

Geometric shapes are observed to perform much better than **winglet**
Smoother curves / taper contribute positively to performance



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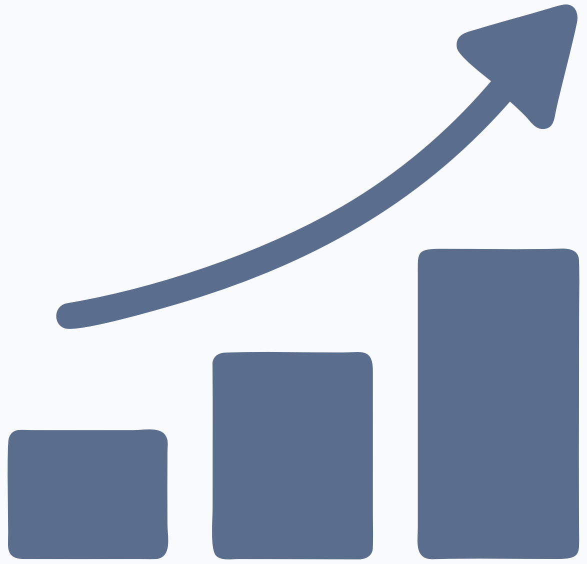
Thickness

Optimal thickness remains **5-10mm**
Some geometric shapes have wider range of performance (Triangular)



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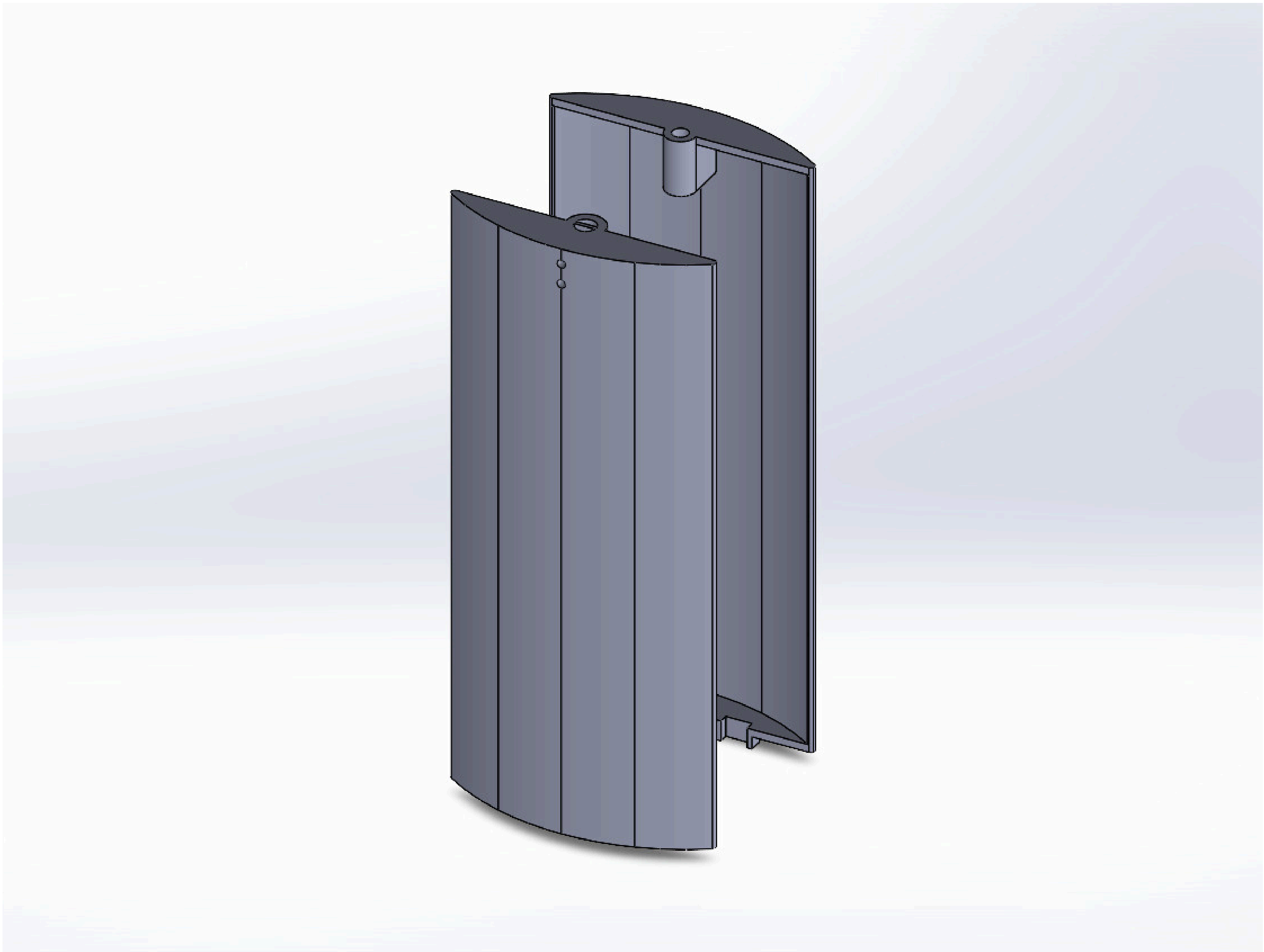
Trends

Generally better performance
Strong inter-correlation of metrics
Highest correlation between **Power & Efficiency**

Best Performing Foil Geometry Overview



Circular Foils (Geometric)

Displayed highset values across **all metrics**
Highest efficiency recorded (~45%)

Conclusion

Conclusions: Overview

Summary of key findings and insights

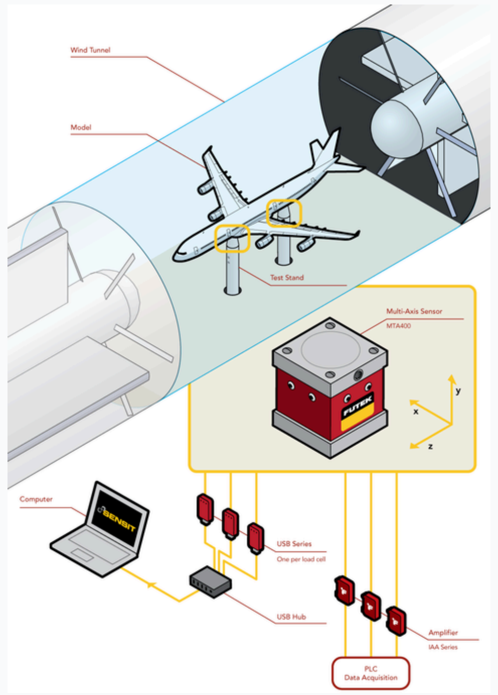
Geometric Influence

Geometry has **strong influence**
Full planform shape perform better
Smoother curves / tapers have positive effects

Thickness Infuence

Most shapes have ideal thicknesses
Optimal range usually **low-moderate** (5-10mm)
Some have a **wider range**, remaining fairly effective at 15mm

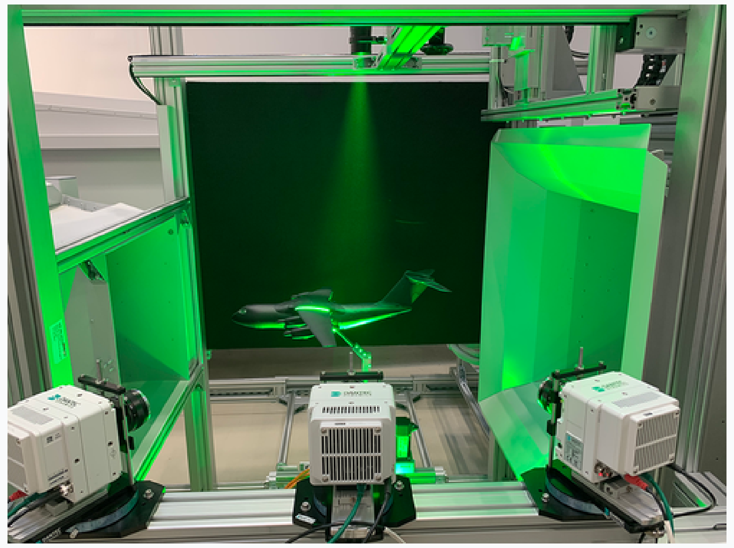
Experimental Limitations



("Futek | Applications | Wind Tunnel Testing")

Loadcells

Allows accurate **force / lift** representation
Assist better in calculations of metrics



("Particle Image Velocimetry | PIV Laser Measurement")

Particle Image Velocimetry

Use of **PIV / dye injection**
Allows **direct visualisation** of vortex
More direct link between performance

Questions & Answers

References

[1] M. Johnston, "10 Biggest Renewable Energy Companies in the World," Investopedia, Jul. 19, 2024. https://www.investopedia.com/investing/top-alternative-energy-companies/ (accessed Dec. 04, 2025).

[2] A. Liebendorfer, "Imitating Fishtail Motion to Increase Efficiency of Flapping Foil Energy Harvesters," Scilight, vol. 2023, no. 31, Aug. 2023, doi: https://doi.org/10.1063/10.0020611.

[3] M. De Manabendra, Y. Sudhakar, S. Gadde, D. Shanmugam, and S. Vengadesan, "Bio-inspired Flapping Wing Aerodynamics: A Review," Journal of the Indian Institute of Science, vol. 104, no. 1, pp. 181–203, Jan. 2024, doi: https://doi.org/10.1007/s41745-024-00420-0.

[4] U. Karthikeyan and J. Hussain, "Enhancing the efficiency of wind energy conversion systems using Novel airfoil based small scale wind turbine," Matéria (Rio de Janeiro), vol. 30, no. 1, Jan. 2025, doi: https://doi.org/10.1590/1517-7076-rmat-2024-0827.

[5] "Futek | Applications | Wind Tunnel Testing," Scigate.com.sg, 2025. https://scigate.com.sg/products/29-futek/4654-applications-wind-tunnel-testing (accessed Dec. 05, 2025).

[6] "Particle Image Velocimetry | PIV Laser Measurement," Dantec Dynamics | Precision Measurement Systems & Sensors. https://www.dantecdynamics.com/solutions/fluid-mechanics/particle-image-velocimetry-piv/ (accessed Dec. 06, 2025).